

you 1m



lab !!

$$\Delta p = F\Delta t = m\Delta v = m(v_f - v_i)$$

$$P = F\Delta t$$

$$P = m\Delta v = m(v_f - v_i)$$

/20

Investigating Impulse Lab

Name: _____

Purpose

Investigate how the size of an impulse, created by different follow-through distances, can affect the final velocity and displacement of an object, which is a basketball in this lab.

Materials

- Basketball
- Measuring Tape
- Ruler
- Stopwatch
- Spring Scale
- Garbage Bag
- Masking Tape



$$m_{\text{ball}} = 0.260$$

$$mgh = \frac{1}{2}mv_f^2$$

$$\frac{2mgh}{m} = v_f^2$$

$$\sqrt{\frac{2(0.260\text{ kg})(9.8\text{ m/s}^2)(1.21\text{ m})}{(0.260\text{ kg})}} = v_f$$

Hypothesis (2 marks)

The greater the arm extension, the greater final velocity is leading to greater impulse $\rightarrow$ displacement

$$v_f =$$

$$\Delta E_m = \frac{1}{2}mv_f^2$$

Procedure

1. Set up the measuring tape by extending it to an appropriate length on a flat surface.
2. Measure and record the mass of the ball with the aid of the garbage bag and spring scale.
3. Measure and record the full extension of the arm length and the initial vertical height of the ball using the meter stick.
4. Stand behind the start of the tape measure where the full extension of the arm will line up to when the measuring tape is at zero.
5. Hold the ball at chest level and without stepping forward, throw the ball horizontally in the forward direction.
6. Record the time of the ball in contact with the person and the time before the basketball hits the ground. Measure and record the horizontal displacement of the ball.
7. Repeat for five trials.
8. Repeat steps 2-3 having only half the extension of the arm for five trials.

Results (3 marks)

Make a table to record results of the lab.

$$v_i = 0\text{ m/s}$$

$$\text{height from hand down} = 121\text{ cm}$$

extension of arm
↓

full of

half ext

distance	distance ball traveled	extension + traveled dis	time	VF
46 cm	240 cm	286 cm	.61 s	
46 cm	248 cm	294 cm	.40 s	
46 cm	245 cm	291 cm	.58 s	
46 cm	231 cm	277 cm	.45 s	
46 cm	247 cm	293 cm	.40 s	
46 cm	247 cm	293 cm	.38 s	
23 cm	210 cm	233 cm	.48 s	
23 cm	218 cm	241 cm	.48 s	
23 cm	197 cm	220 cm	.47 s	
23 cm	194 cm	217 cm	.28 s	
23 cm	207 cm	230 cm	.40 s	
23 cm			.31 s	



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ou 50s



Graph/Calculations: (5 marks)

1. Use the data collected to calculate the final velocity of the basketball when it was thrown. Make sure to analyze if the velocity and displacement of the basketball changed when the arm had different lengths of extension. (2 marks)

full extension:

$$\frac{d}{t} = \frac{v}{t}$$

$$v = \frac{0.46m}{0.136s}$$

$$V_f = 3.38m/s \text{ [fwd]}$$

half extension:

$$\frac{d}{t} = \frac{v}{t}$$

$$v = \frac{0.23m}{0.101s}$$

$$V_f = 2.30m/s \text{ [fwd]}$$

2. Use the data collected to calculate the impulse of the basketball in both situations. (3 marks)

full ext:

$$\Delta p = m \Delta v$$

$$= (0.260 \text{ kg}) (3.38m/s)$$

$$\Delta p = 0.879 \text{ N}\cdot\text{s}$$

half ext:

$$\Delta p = m \Delta v$$

$$= (0.260 \text{ kg}) (2.3m/s)$$

$$\Delta p = 0.598 \text{ N}\cdot\text{s}$$

3. Make a graph showing the relationship between the impulse and the magnitude of the final velocity of the ball in both situations. ***

Analysis (10 marks)

1. Beginning with reference to your hypothesis, discuss whether your hypothesis was correct. If correct, support with reference to your data and calculations. Make sure if you were not correct to use data and calculations to establish a more accurate one. (3 marks)
2. Using your knowledge gathered from the lab, explain why airbags are used in cars as opposed to "brick bags." (2 marks)
3. Discuss and explain possible sources of error and uncertainty in the experimental design and measurement of the lab. (2 marks)
4. Make sure to finish with a conclusion; that is, what relationship have you discovered between the impulse of an object and its final velocity and displacement that can be made into a rule or general statement. (3 marks)



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